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3rd Sem / Civil, Highway Engg.
Subject : Structural Mechanics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Bulk modulus is measured in terms of (CO1)

- a) N/m b) N/m^2
- c) Nm/s d) Ns/m^2

Q.2 Maximum bending moment occurs at a point where (CO2)

- a) Shear force is maximum
- b) Transverse loading is zero
- c) Shear force is either zero or changes sign
- d) None

Q.3 A cantilever is a beam whose? (CO2)

- a) One end is fixed and other is free

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Q.11 The strength of a beam mainly depends on _____
(CO4)

Q.12 A frame in which $n=2j-3$ is called _____. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the proportional limit, elastic limit and yield point in tensile test of mild steel. (CO1)

Q.14 Explain the types of stresses and strains. (CO1)

Q.15 Draw shear force and bending moment diagram for a simply supported beam carrying concentrated load W KN at mid-point of the span? (CO2)

Q.16 A cantilever beam 3 m long carries loads of 2 KN and 1.5 KN at a distance of 1 m and 2 m from free end respectively. Draw S.F and B.M diagrams. (CO2)

Q.17 State and prove the theorem of parallel axis. (CO3)

Q.18 The diameter of a circular section is 100 mm, find moment of inertia about horizontal axis passing through centroid. (CO3)

Q.19 List the assumptions made in the theory of simple bending. (CO4)

Q.20 Determine the maximum slope and deflection in a cantilever beam carrying a point load at the free end. (CO4)

Q.21 Write down the assumptions and limitations of Euler's formula. (CO5)

Q.22 Explain method of joints for analysis of truss. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 A crane lifts a load of 30 KN. Find the diameter of the steel wire of the crane, so that the stresses produced in the wire does not exceed 100 N/mm^2 . Also find the elongation per meter length of the rope when $E = 2 \times 10^5 \text{ N/mm}^2$. (CO1)

Q.24 Explain types of end supports and types of beams with neat sketches. (CO2)

Q.25 Prove the bending equation i.e. $M/I = \sigma/y = E/R$ (CO4)